



NANYANG TECHNOLOGICAL UNIVERSITY

EE6407 Genetic Algorithms & Machine Learning

Assignment 1 & 2 Report

MSc in Electrical and Electronic Engineering

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1 Introduction

For the course EE6407, I implement two assignments, including the Naïve Bayes Classifier and the Fisher Linear Discriminate Classifier. This report summarizes the methods and results of these two assignments. I use Python as the programming language and Jupyter Notebook as the development environment. The followings show the details of my design and implementation.

2 Assignment 1

2.1 Handling Missing Values and Outliers

2.1.1 Missing values

Detection:

For the missing value detection, I use the pandas library to check, with `isna()` function. This function would return a matrix reflects if the content in each position is empty or not. By summing the values in each column, I can determine the number of times each feature's missing values occur 1.

```
missing_train=Train_data.isna()  
missing_train_n=missing_train.sum()
```

Figure 1: Missing Value Detection Code

At first, there seems no missing values. however, after trying to transform data type from `dtype` to `float`, there seems several data cannot successfully transformed, leading to new missing values. The missing value detection for training set consequence 1:

Feature	Missing Value Count
A	2
B	0
C	0
D	1
E	0

Table 1: Missing Value Count

Handling:

Since there are only several missing values in each feature, eg. two for feature A and one for feature D. I think that there are three methods could be used to replace the empty with practical values [1]:

- ***Missing Deletion:***

Simply delete all the feature at the position of missing value. This method is simple and effective, but may lead to data loss and bias of training classifier. In this condition, since the total number of missing values is 3, which is 2.5% of the total data, I think this method may be acceptable.

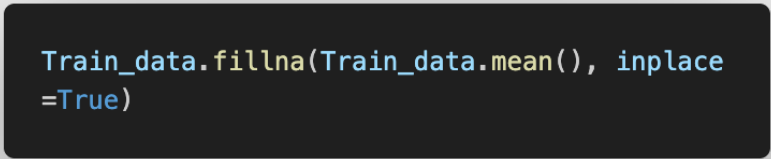
- ***Single Imputation:***

Replace the missing value with a specific value, such as mean, median, or k nearest neighbors. This method avoid the data loss, and consider the distribution of data. However, when there are plenty of missing value, this method may replace these values with same data, leading to bias.

- ***Imputation with Machine Learning:***

Use machine learning methods, regression, to predict the missing value. For example, use linear regression, decision tree regression, or least square regression to predict. Compared with single imputation, this method is more reliable. But it is more complex and time-consuming.

In this assignment, I use the mean value of each feature to replace the missing values. Because the number of missing values is small, I think there is no need to use more complex methods, like machine learning. And this implementation is more practical than deletion 2.



```
Train_data.fillna(Train_data.mean(), inplace=True)
```

Figure 2: Imputation of Missing Values

2.1.2 Outliers

Detection:

abcd

Handling:

2.2 Design a Naïve Bayes classifier

```
class GNB_Classifier:
    def __init__(self):
        pass

    def fit(self,X,y):
        self.classes=np.unique(y)
        self.means=[]
        self.vars=[]
        self.priors=[]
        for w in self.classes:
            X_w=X[y==w]
            prior_w=X_w.shape[0]/X.shape[0]
            mean_w=np.mean(X_w,axis=0)

            # cov_w=np.cov(X_w,rowvar=False)+1e-6*np.eye
            (X.shape[1])
            var_w = np.var(X_w, axis=0)
            self.priors.append(prior_w)
            self.means.append(mean_w)
            # self.vars.append(cov_w)
            self.vars.append(var_w)

        self.priors = np.array(self.priors)
        self.means = np.array(self.means)
        self.vars = np.array(self.vars)

    def ConditionPwithGaussian(self,x,mean,
var):
        eps = 1e-6
        coeff = 1.0 / np.sqrt(2 * np.pi *
var + eps)
        exponent = - (x - mean) ** 2 / 2 *
var + eps)
        return coeff * np.exp(exponent)
```

```

def predict(self, X):
    X = np.array(X)
    y_pred = []

    for x in X:
        posteriors = []
        for i, c in enumerate(self.classes):
            prior_log = np.log(self.priors[i])

            likelihood_log = np.sum(np.log(
self.ConditionPwithGaussian(x, self.means[i]
], self.vars[i])))
            posteriors.append(prior_log +
likelihood_log)
            y_pred.append(self.classes[np.argmax
(posterior)])

        return np.array(y_pred)

def evaluate(self, X, y):
    y = np.array(y)
    y_pred = self.predict(X)
    accuracy = np.mean(y_pred == y)
    return accuracy

```

Figure 3: Code of Classifier Establishment

2.3 Prediction and Analysis

Table 2: Results of Gaussian Naive Bayes Classifier

Metric	Value
Training Accuracy	0.958
Predicted Labels	[1 1 1 1 1 1 1 1 1 3 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3]

3 Assignment 2

3.1 Fisher Linear Discriminate Classifier

Establish:

Training:

3.2 Prediction and Analysis

Table 3: Results of Fisher Linear Discriminant Analysis (LDA)

Metric	Value
Predicted Labels	-1 -1 -1 -1 -1 -1 -1 -1 -1 1 1 1 1 1 1 1 1 1 1 1
$\mathbf{w}$	0.001786 0.003473 0.001947 0.001683 0.002658
b	-0.008700
Training Accuracy	0.932

4 Conclusion

References

- [1] L. Ren, T. Wang, A. Sekhari Seklouli, H. Zhang, and A. Bouras, “A review on missing values for main challenges and methods,” *Information Systems*, vol. 119, p. 102268, Oct. 2023.